The Bacterial Glutamate-to-P5C Branch of Proline Biosynthesis: ProB and ProA

A commissioned review synthesizing the two-enzyme module that converts L-glutamate to Δ¹-pyrroline-5-carboxylate (P5C)

1. Executive summary

In bacteria, the first committed leg of proline biosynthesis converts the abundant, central-metabolite amino acid L-glutamate into 1-pyrroline-5-carboxylate (P5C) in two enzymatic steps, catalyzed by **ProB** (glutamate 5-kinase, G5K/GK; EC 2.7.2.11) and **ProA** (γ-glutamyl phosphate reductase, GPR; EC 1.2.1.41). ProB uses ATP to phosphorylate the γ-carboxylate of glutamate, producing the acyl-phosphate **L-glutamyl 5-phosphate (γ-glutamyl phosphate, GP)**. ProA then reductively dephosphorylates this intermediate using **NADPH**, releasing inorganic phosphate and generating **L-glutamate 5-semialdehyde (GSA)**, which spontaneously and reversibly cyclizes to P5C. This module is the flux-controlling entry into proline (and, indirectly, ornithine/arginine) metabolism and is the principal site of end-product feedback control.

Four features dominate the biology of the module. First, ProB is a two-domain (AAK + PUA) allosterically regulated enzyme that is **feedback-inhibited by proline** (and, in some organisms, ornithine), making it the committed and rate-controlling step (Marco-Marín et al., 2007, P 17321544). Second, the intermediate GP is **chemically labile**, which imposes a physical requirement for tight functional coupling — probably substrate channeling — between ProB and ProA (Arentson et al., 2012, P 22201749). Third, the module is deeply conserved: the same two activities are welded into a single bifunctional enzyme, **Δ¹-pyrroline-5-carboxylate synthase (P5CS)**, in most eukaryotes, where the N- and C-terminal domains correspond directly to bacterial ProB and ProA (Hu et al., 1992,

).

P 1384052

Fourth, both enzymes belong to **ancient, pan-domain superfamilies** — ProB to the amino-acid-kinase (AAK) family (with ArgB, carbamate kinase, aspartokinase) and ProA to the acyl-phosphate–reducing semialdehyde-dehydrogenase family (with ArgC and ASADH) — so the module is best understood as one specialization of a broadly reused "kinase-then-NADPH-reductase" toolkit shared with arginine and aspartate-family biosynthesis (McLoughlin & Copley, 2008,

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P 18757760

The scope defined here deliberately **stops at P5C**, before the terminal, NAD(P)H-dependent P5C reductase (ProC) step that produces L-proline.

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2. Definition and biological boundaries

2.1 What is included

The module comprises exactly two catalytic activities and one non-enzymatic conversion:

Step	Enzyme (gene)	Reaction	Cofactor
1	Glutamate 5-kinase (**proB**)	L-glutamate + ATP → L-glutamyl 5-phosphate + ADP	ATP, Mg^{2+}
2	γ-Glutamyl phosphate reductase (**proA**)	L-glutamyl 5-phosphate + NADPH → L-glutamate-5-semialdehyde + P_i + $NADP^+$	NADPH
(spontaneous)	—	L-glutamate-5-semialdehyde ⇌ P5C + H_2O	none

The gene-to-enzyme assignments (proB → kinase, proA → reductase) were established biochemically and genetically in *E. coli* (Hayzer & Leisinger, 1980,

;

P 6255065

Deutch et al., 1984,

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P 6089111

2.2 What should be treated separately (adjacent processes often conflated)

- **The ProC / P5C reductase step.** P5C reductase (ProC) reduces P5C to proline using NAD(P)H. It is a distinct enzyme with a separate gene and regulation, and is explicitly *outside* this module's boundary. The module's product P5C is the substrate for ProC.
- **Proline catabolism (PutA / PRODH + P5CDH).** The same intermediates (P5C/GSA, glutamate) appear in the *degradative* direction, where proline dehydrogenase and P5C dehydrogenase (often fused as PutA) run oxidatively toward glutamate. This is a chemically distinct, oppositely directed pathway; the GSA↔P5C node is shared, but the enzymes, cofactor logic (FAD/NAD^+ vs. ATP/NADPH), and physiology differ (Korasick et al., 2017, P 29295473).
- **The ornithine/arginine branch.** GSA/P5C is a branch point shared with ornithine biosynthesis: ornithine aminotransferase interconverts GSA and ornithine. In organisms where P5CS/ProAB feed this node, the module also supports arginine synthesis. This shared node — not the ProB/ProA chemistry itself — is what links proline to the urea-cycle-adjacent network in mammals (Pérez-Arellano et al., 2010, P 20091669).
- **N-acetylglutamate kinase (NAGK, ArgB) and N-acetylglutamyl-phosphate reductase (ArgC).** These arginine-pathway enzymes are the closest structural/mechanistic homologs of ProB and ProA, respectively (same AAK and aldehyde/acyl-phosphate-reductase families), and are frequently confused with them. They act on *N-acetylated* glutamate substrates and belong to arginine biosynthesis. Their homology is central to the evolutionary story (Section 4) but they are not part of this module (McLoughlin & Copley, 2008, P 18757760).
- **Osmoadaptive vs. anabolic proline synthesis.** In some bacteria (e.g., *Bacillus subtilis*) a paralogous, stress-inducible kinase (ProJ) substitutes for ProB; this is a regulatory variant of the same chemistry (Section 5), not a separate pathway type.

2.3 Competing definitions

There is little genuine controversy over the module's chemistry, but two definitional ambiguities recur in the literature. (i) Some authors describe "the first two steps of proline biosynthesis" as producing "GSA" and others as producing "P5C"; these are the same thing because GSA and P5C are in spontaneous, pH-dependent equilibrium. (ii) In the eukaryotic literature the two steps are almost always discussed as a *single* enzyme (P5CS), which can obscure the fact that they are two chemically independent reactions with a discrete, isolable (if unstable) intermediate. This review treats the bacterial two-gene arrangement as the reference architecture.

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3. Mechanistic overview

3.1 Step 1 — ProB: glutamate phosphorylation (committed step)

ProB transfers the γ-phosphate of ATP to the δ-(γ-)carboxyl oxygen of L-glutamate, forming the mixed acyl-phosphate anhydride L-glutamyl 5-phosphate and ADP. The reaction requires Mg^{2+} and, notably, free Mg^{2+} beyond the ATP-chelated pool (Marco-Marín et al., 2007, P 17321544).

This is the **committed and rate-controlling step**: it is the point of allosteric feedback inhibition by the end product proline (and ornithine in organisms with a combined route), and essentially all proline-overproduction / osmotolerance mutants that bypass end-product control map to *proB* (Section 5.3).

Mechanistically ProB belongs to the **amino acid kinase (AAK) superfamily** of acyl-phosphate-forming enzymes, the same family as carbamate kinase and N-acetylglutamate kinase. Catalysis occurs in a crater on the C-edge of the AAK β-sheet; crystal structures captured bound glutamate, the product GP, and the proline analogue 5-oxoproline in the regulatory/active region (Marco-Marín et al., 2007, P 17321544).

3.2 Step 2 — ProA: reductive dephosphorylation to GSA/P5C

ProA catalyzes an NADPH-dependent reductive dephosphorylation of the acyl phosphate. Mechanistically it belongs to the **acyl-phosphate–reducing "semialdehyde dehydrogenase" family** (the ArgC / aspartate-β-semialdehyde-dehydrogenase, ASADH, class), *not* the oxidative ALDH superfamily. By close analogy to ASADH, a catalytic active-site cysteine attacks the acyl-phosphate carbonyl to expel inorganic phosphate and form a covalent thioacyl-enzyme intermediate; NADPH then reduces this thioacyl to the aldehyde, releasing L-glutamate-5-

semialdehyde (Section 4.2). The nascent GSA is in spontaneous equilibrium with its cyclized form, P5C, via intramolecular Schiff-base formation between the aldehyde and the α-amino group.

3.3 The intermediate problem and channeling

The defining physical constraint of the module is that **γ-glutamyl phosphate is extremely labile**: as a reactive acyl phosphate it hydrolyzes rapidly and can cyclize non-productively to 5-oxoproline (pyroglutamate) if released into bulk solvent. ProB itself is described as making a "highly unstable product" (Marco-Marín et al., 2007, P 17321544).

The accepted rationale is therefore that GP must be handed from ProB to ProA with minimal solvent exposure: "Substrate channeling of gamma-glutamyl phosphate is thought to be necessary to protect it from bulk solvent" (Arentson et al., 2012, P 22201749).

This constraint provides a unifying explanation for three otherwise separate observations: (i) the genetic linkage of *proB* and *proA* in a *proBA* operon in many bacteria; (ii) the physical fusion of the two activities into one polypeptide (P5CS) in eukaryotes; and (iii) the difficulty of assaying GP as a free intermediate.

Two caveats are important for rigor. First, *direct* structural proof of a ProB→ProA channel (e.g., a solved ProB·ProA complex with an internal tunnel) is, to the best of the surveyed literature, **weaker than for the catabolic PutA channel**, where P5C/GSA channeling between PRODH and P5CDH active sites is structurally and kinetically established (Arentson et al., 2012, P 22201749).

Channeling in the *biosynthetic* direction is strongly inferred from chemistry and genetics but remains partly a working model. Second, in eukaryotic P5CS the intramolecular hand-off between the fused GK and GPR domains is the natural structural embodiment of this coupling.

3.4 Obligatory, conditional, and accessory features

- **Obligatory:** ATP/Mg^{2+} for step 1; NADPH for step 2; the strict order (phosphorylation must precede reduction — ProA cannot act on glutamate directly); protection/rapid consumption of GP.

- **Conditional:** the identity of the kinase (ProB vs. the osmostress paralog ProJ); operon organization (linked *proBA* vs. dispersed genes); sensitivity to ornithine (present in "combined-route" organisms/eukaryotes, weaker or absent in others).
- **Accessory:** the PUA domain of ProB (see Section 4) modulates oligomerization/regulation but is not catalytic; downstream ProC/ProI reductases and ornithine aminotransferase lie outside the module but determine the fate of P5C.

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4. Major molecular players and active assemblies

4.1 ProB / glutamate 5-kinase — a novel two-domain architecture

The *E. coli* G5K crystal structures revealed a previously unseen architecture within the AAK family: each subunit is built from **a 257-residue N-terminal AAK domain** ($\alpha_3\beta_8\alpha_4$ sandwich; catalysis + proline inhibition) and **a 93-residue C-terminal PUA domain** ($\beta_5\beta_4$ sandwich plus three α-helices; a fold typical of RNA-modifying enzymes) (Marco-Marín et al., 2007, P 17321544).

Functionally significant points:

- **Oligomerization.** G5K is a **tetramer (dimer of dimers)**, confirmed independently by cross-linking of the recombinant enzyme, which purifies in "the highly active proline-inhibitable form" (Pérez-Arellano et al., 2004, P 15502337). The tetramer is flat and elongated; the two dimers interact exclusively through their AAK domains, presenting two exposed active centres per face.
- **Regulatory site.** Feedback inhibitor (proline / the analogue 5-oxoproline) binds in the AAK-domain crater, coupling end-product sensing to catalysis. The PUA domain and inter-subunit contacts are implicated in transmitting/regulating this inhibition.
- **Feedback-resistance mutations cluster.** Independent proline-overproducer mutants map to a short conserved region of GK — e.g., a conserved ~26-amino-acid segment in *Listeria* (Sleator et al., 2001, P 11571156) and the Ala117→Val change in *Serratia* that lowers proline sensitivity 700-fold (Omori et al.,

1992,

P 1316937

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4.2 ProA / γ-glutamyl phosphate reductase — an acyl-phosphate-reducing semialdehyde dehydrogenase and a shared node

Family and mechanism. ProA is best classified not with the ALDH (aldehyde→acid oxidative) superfamily but with the **NADP-dependent, acyl-phosphate–reducing "semialdehyde dehydrogenase" family**, whose paradigmatic members are N-acetyl-γ-glutamyl-phosphate reductase (**ArgC**, arginine pathway) and aspartate-β-semialdehyde dehydrogenase (**ASADH**, aspartate pathway). These enzymes carry out the same chemistry ProA performs — reductive dephosphorylation of an acyl phosphate to a semialdehyde — via a conserved **catalytic cysteine that forms a covalent thioacyl-enzyme intermediate**, assisted by a His general base and Arg/Glu substrate-positioning residues, with NADPH delivering the hydride. This mechanism is directly established for ASADH: active-site Cys (Cys130 in *M. tuberculosis*, Cys136 in *H. influenzae*) has been trapped covalently and mutation to Ser collapses activity (Blanco et al., 2004,

P 15388927

; Vyas et al., 2012,

P 22683789

).
Importantly, **the fine catalytic residues of bacterial ProA are inferred by homology from ArgC/ASADH**, because a solved bacterial ProA structure is not, to the best of this survey, available — a genuine gap (Section 7). This family assignment is *not* interchangeable with the ALDH superfamily that runs the opposite, oxidative reaction in proline/glutamate catabolism (GSALDH/P5CDH).

Two further biologically important properties recur:

- **Promiscuity toward the arginine pathway.** Wild-type *E. coli* ProA has a low promiscuous activity on N-acetylglutamyl phosphate (the ArgC substrate). A single active-site mutation (Glu383→Ala) raises this promiscuous N-acetylglutamyl-phosphate reductase activity ~12-fold while collapsing the native activity ~2,800-fold — a clean demonstration of the mechanistic kinship between ProA and ArgC and of how enzyme "gene sharing" could seed evolution of a new activity (McLoughlin & Copley, 2008,

  P 18757760

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- **A convergence point for parallel routes.** In *B. subtilis*, ProA serves *both* the anabolic (ProB–ProA–ProI) and the osmostress (ProJ–ProA–ProH) proline routes; consequently *proA* deletion is a proline auxotroph and simultaneously abolishes osmoadaptive proline production, whereas deletion of the stress-specific genes does not (Brill et al., 2011, P 21784929; Hoffmann et al., 2017, P 28752945).

4.3 The bifunctional eukaryotic counterpart (P5CS)

Plants, animals, and fungi generally fuse the two activities. The mothbean P5CS cDNA encodes a single polypeptide with both GK and GSA-dehydrogenase activities whose two domains "correspond to the ProB and ProA proteins of *Escherichia coli*" (Hu et al., 1992, P 1384052). Human P5CS is a mitochondrial, ATP- and NAD(P)H-dependent bifunctional enzyme; its loss causes an inborn error with hyperammonemia, hypoornithinemia, hypocitrullinemia, hypoargininemia and hypoprolinemia (Baumgartner et al., 2000, P 11092761), and isoform choice (a 2-aa insert at the GK domain) tunes ornithine feedback sensitivity and tissue role (Hu et al., 2008, P 18401542).

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5. Evolutionary and cell-biological variation

5.1 Split genes vs. fused enzyme across lineages

The clearest phylogenetic pattern is architectural. **In prokaryotes, fungi, and some unicellular eukaryotes (green algae), the first two reactions are carried out by two distinct enzymes — GK (proB) and GPR (proA)**, whereas most eukaryotes use a single bifunctional P5CS (Filgueiras et al., 2024, P 38693917).

The bacterial two-gene state is ancestral; eukaryotic fusion is a derived elaboration that hard-wires the ProB→ProA coupling and folds proline synthesis into the shared glutamate/ornithine/proline interconversion network (Pérez-Arellano et al., 2010, P 20091669).

5.1b Deepest plausible origin, and which parts are ancient vs. derived

The module's two catalytic activities descend from **two independently ancient enzyme superfamilies**, and this is the strongest clue to its origin:

- **The kinase (ProB)** belongs to the **amino acid kinase (AAK) superfamily** of carboxylate-phosphorylating (acyl-phosphate-forming) enzymes, whose members — carbamate kinase, N-acetylglutamate kinase (ArgB), aspartokinase, UMP kinase — are distributed across Bacteria, Archaea, and Eukarya. A phosphoryl-transfer AAK core is therefore plausibly a component of very early (LUCA-adjacent) amino-acid metabolism (Marco-Marín et al., 2007, P 17321544).
- **The reductase (ProA)** belongs to the equally widespread **acyl-phosphate–reducing semialdehyde-dehydrogenase family** (ArgC, ASADH), which supplies semialdehyde precursors to arginine, lysine/threonine/methionine (the aspartate pathway) and proline. Its deep conservation across prokaryotes mirrors that of the AAK kinases (McLoughlin & Copley, 2008, P 18757760).

The recurring "**kinase-activates-carboxylate → NADPH-reduces-acyl-phosphate-to-semialdehyde**" logic is shared by at least three biosynthetic branches (proline: ProB/ProA; arginine: ArgB/ArgC; aspartate family: aspartokinase/ASADH). The most parsimonious reading is that an **ancestral, comparatively substrate-tolerant kinase+reductase pair** was duplicated and specialized, with the acetylated (arginine) and non-acetylated (proline) glutamate branches diverging by gain/loss of the N-acetyl handle. On this view:

- **Ancient/conserved:** the AAK phosphoryl-transfer core of ProB and the Cys-thioacyl reductase core of ProA; the obligatory kinase-then-reductase order; NADPH dependence; the labile acyl-phosphate intermediate.
- **Later elaborations / lineage-specific:** the **C-terminal PUA regulatory domain** and proline-feedback site of ProB (a regulatory add-on to the catalytic AAK core); **operon linkage** (present in enterobacteria/*Thermus*, absent in *Campylobacter*); the **osmostress**

paralog ProJ and stress operon *proHJ*; and the **eukaryotic ProB–ProA gene fusion** into P5CS with mitochondrial targeting.

- **Best representative of the ancestral role:** because the family has expanded, the *generalist* behavior is best seen in enzymes retaining cross-reactivity — e.g., wild-type **ProA's residual ArgC activity** is a direct experimental window onto the pre-specialization, substrate-tolerant ancestor (McLoughlin & Copley, 2008, P 18757760). For the kinase, **ArgB/N-acetylglutamate kinase** is the closest well-characterized structural relative and a useful proxy for the ancestral AAK fold before the PUA regulatory elaboration.

Uncertainty: precise gene-tree topologies, the exact order of duplication/fusion events, and whether the proline or arginine branch is the more ancestral use of the shared toolkit remain unresolved; the 2024 phylogenetic synthesis supports an ancestral two-gene state with independent eukaryotic fusion but does not settle these deeper node orderings (Filgueiras et al., 2024, P 38693917).

5.2 Operon organization varies even within bacteria

- **Linked *proBA* operon:** *E. coli* (proB precedes proA; the two genes are contiguous and co-transcribed) (Deutch et al., 1984, P 6089111) and *Serratia marcescens* (Omori et al., 1991, P 1851803). In *Thermus thermophilus* the two genes overlap by 2 bp within one operon (Kosuge et al., 1994, P 7988899).
- **Unlinked genes:** In *Campylobacter jejuni* the proline biosynthetic genes are **not** tandemly arranged; proA is dispersed and expressed from its own promoter (Louie & Chan, 1993, P 8341262). Thus operon linkage is a common but not universal solution to co-expressing the coupled enzymes.

5.3 Regulatory and physiological states: anabolic vs. osmoadaptive proline

Proline doubles as a **compatible solute**. Many bacteria massively upregulate proline synthesis under hyperosmotic stress, and the module is embedded in this stress physiology:

- ***B. subtilis* uses a duplicated kinase.** Housekeeping proline uses ProB; osmoadaptive proline uses the paralog **ProJ**, together with the shared ProA and a stress-specific P5C reductase ProH. The **proHJ operon is osmotically inducible via a SigA-type promoter, whereas proBA is constitutive** (Brill et al., 2011, P 21784929). Single-cell studies show heterogeneous, switch-like proHJ activation on osmotic upshift (Morawska et al., 2022, P 35593133), and metabolic flux is routed toward the glutamate precursor at the 2-oxoglutarate node during salt stress (Kohlstedt et al., 2014, P 24571712).
- **Feedback control is the master switch.** Because GK is proline-feedback-inhibited, osmotolerant/proline-overproducing phenotypes are routinely obtained by desensitizing GK — via point mutations in *proB* (*Serratia* A117V, ~700-fold desensitized; *Thermus*, *Listeria*, *E. coli proBosm*) (Omori et al., 1992, P 1316937; Kosuge & Hoshino, 1998, P 9797285; Sleator et al., 2001, P 11571156). Heterologous expression of desensitized *proBA* raises intracellular proline and osmotolerance across taxa (Rhizobium, transgenic Arabidopsis) (Neumivakin et al., 1995, P 8554600; Chen et al., 2007, P 17562291).

5.4 Where control is exerted: enzyme, not operon

A point of practical importance is *where* flux is regulated. In enterobacteria the module is controlled principally **post-translationally, at the ProB step**, by allosteric feedback inhibition of GK. The *proBA* operon itself is **not** subject to proline-mediated transcriptional feedback repression (shown with *lacZ–proBA* fusions in *Serratia*), and — unlike amino-acid operons such as *ilvGMEDA* — it lacks a leader-peptide attenuation control (Omori et al., 1991, P 1851803). This division of labor (fast allosteric switch on the committed enzyme; largely constitutive transcription) suits proline's dual identity as a proteinogenic amino acid and a rapidly deployable compatible solute — contrast the separately, osmotically induced *proHJ* route (Section 5.3). By comparison, proline *supply* is used elsewhere as a signal: *Salmonella* reads out proline-charged $tRNA^{Pro}$ through *mgtCBR* leader attenuation, and proline auxotrophs are defective for intramacrophage survival and murine virulence in a host niche that is both proline-limited and osmotically stressful (Lee et al., 2014, P 24516160). The module's output therefore couples to physiology (osmoadaptation, host colonization) that in turn shapes demand.

5.5 Compartmentalization in eukaryotes

Whereas the bacterial module is cytoplasmic, the fused eukaryotic P5CS is **mitochondrial** in mammals (Baumgartner et al., 2000, P 11092761) and its localization/isoform usage couples it to organelle redox and to ornithine/arginine metabolism — a cell-biological layer that has no direct counterpart in bacteria.

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6. Constraints, dependencies, and failure modes

- **Strict reaction order.** Phosphorylation must precede reduction: ProA cannot reduce glutamate directly — it requires the activated acyl phosphate. There is no known bacterial route from glutamate to GSA that bypasses the GP intermediate.

- **Cofactor specificity.** Step 1 needs ATP + Mg^{2+} (including free Mg^{2+}); step 2 needs NADPH specifically. This ties module flux to the cell's energy charge and NADPH/$NADP^+$ ratio.
- **Intermediate lability enforces coupling.** GP's short half-life means the two steps cannot be arbitrarily separated in space or time; this is the physical basis for operon linkage, gene fusion, and probable channeling (Section 3.3). Loss of coupling is a plausible failure mode that would divert GP to pyroglutamate.
- **Mutually exclusive directionality.** The biosynthetic module (ATP/NADPH-consuming, glutamate→P5C) and proline catabolism (FAD/NAD^+, proline→glutamate) share intermediates but are thermodynamically and enzymatically opposed; futile cycling is avoided by separate enzymes and regulation.
- **Branch-point competition.** GSA/P5C is contested between ProC (→proline) and ornithine aminotransferase (→ornithine/arginine). The partition is set downstream of this module, but end-product feedback on ProB integrates demand from both branches (proline; ornithine in combined-route organisms).
- **Auxotrophy and pleiotropy on failure.** *proA* or *proB* loss produces proline auxotrophy; in *B. subtilis* proA loss additionally abolishes osmostress proline and impairs high-osmolarity growth (Brill et al., 2011, P 21784929). In humans the homologous P5CS deficiency is multisystem (neurodegeneration, connective-tissue and metabolic phenotypes) because it starves the shared proline/ornithine/arginine network (Baumgartner et al., 2000, P 11092761).
- **Host relevance of biosynthetic capacity.** Because host niches can be proline-poor, an intact proline-biosynthesis pathway is often needed for virulence. A *Mycobacterium tuberculosis* proline auxotroph (a *proC*/P5C-reductase mutant — one step downstream of this module) is rapidly killed in macrophages and strongly attenuated in mice, yet protective as a live vaccine (Smith et al., 2001, P 11160012). This underscores that although the module's boundary is at P5C, its output feeds a downstream capacity that pathogens depend on in vivo.

7. Controversies and open questions

1. **Is there a genuine ProB→ProA channel in bacteria?** Channeling is strongly inferred from GP's instability and from the tight genetic/physical coupling, but direct structural evidence (an isolated ProB·ProA complex with a demonstrated tunnel and transfer kinetics) is far less developed than for the *catabolic* PutA channel. Whether bacterial ProB and ProA form a stable complex, a transient encounter complex, or simply operate at high local concentration remains only partly resolved (Arentson et al., 2012, P 22201749).
2. **Mechanistic detail of ProA.** ProA has been far less structurally characterized than ProB; the precise geometry of NADPH binding, the catalytic cysteine/thioacyl chemistry (by analogy to ArgC/GAPDH), and how phosphate release is orchestrated are not as firmly established for the bacterial enzyme as one would like. Much mechanistic inference is imported from homologs (ArgC) and from eukaryotic P5CS.
3. **How feedback signal is transmitted within the ProB tetramer.** The proline site is in the AAK domain, but the roles of the PUA domain, of inter-dimer contacts, and of Mg^{2+} occupancy in propagating inhibition are still model-dependent (Marco-Marín et al., 2007, P 17321544).
4. **Effector spectrum across taxa.** Whether the physiological inhibitor is proline alone or also ornithine varies with the "combined route" status of the organism; mixing data from *E. coli* (proline-inhibited GK) with mammalian P5CS (ornithine-sensitive isoforms) risks overgeneralization. Effector sensitivity should be treated as lineage-specific.
5. **Evolutionary sequence of split→fusion and of ProB/ProJ duplication.** The 2024 phylogenetic synthesis supports an ancestral two-gene state with independent eukaryotic fusion, but the exact number and timing of GK/GPR duplications, fusions, and losses across lineages (and the origin of osmostress paralogs like ProJ) remain active questions (Filgueiras et al., 2024, P 38693917).
6. **Which family member best represents the ancestral role.** Because ProB/ArgB (AAK family) and ProA/ArgC (acyl-phosphate–reducing semialdehyde dehydrogenase family, alongside ASADH) are pairwise homologous, the "ancestral" acyl-phosphate/reductase chemistry is best studied by comparing the non-acetylating (proline) and acetylating (arginine) branches together; the promiscuity experiments on ProA are the clearest window

onto this ancestral, generalist activity (McLoughlin & Copley, 2008,

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Note on scope and uncertainty

This review is bounded at P5C and does not evaluate the ProC/P5C-reductase step. Claims about substrate channeling in the *biosynthetic* direction and about detailed ProA catalytic chemistry rest partly on inference from homologs (ArgC, PutA) and from eukaryotic P5CS, and are flagged as such. Effector sensitivity (proline vs. ornithine) and operon organization are lineage-specific and should not be generalized from any single organism.
